

```
In [1]: import sys, os
sys.path.insert(0, os.path.abspath('.'))

import torch
from src.data.data_loader import SatelliteDataLoader
from src.models.satellite_cnn import SatelliteCNN
from src.models.trainer import SatelliteCNNTrainer

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Using device: {device}')
```

Using device: cuda

```
In [2]: loader = SatelliteDataLoader(
    data_path = "../src/data/UCMerced_LandUse/Images",
    batch_size=32
)

train_loader, val_loader = loader.get_loaders()
class_names = loader.get_class_names()

print(f'Classes: {len(class_names)}\nTraining batches: {len(train_loader)}\nValidat
```

Classes: 21
Training batches: 53
Validation batches: 14

```
In [3]: model = SatelliteCNN(num_classes=len(class_names))
print(f'Model created: {sum(p.numel() for p in model.parameters() if p.requires_gra
```

Model created: 24567893 trainable parameters

```
In [4]: trainer = SatelliteCNNTrainer(model, train_loader, val_loader, device=device)
results = trainer.train(epochs=2)

print(f'\nTest training completed with validation acc: {results["val_accs"][-1]:.2f
```

Training on device: cuda
Training samples: 1680
Validation samples: 420

Epoch 1/2

Training: 100%|██████████| 53/53 [00:32<00:00, 1.64it/s, Loss=2.0536, Accuracy=48.75%]

Validating: 100%|██████████| 14/14 [00:17<00:00, 1.22s/it]

Train Loss: 2.0536, Train Accuracy: 48.75%

Val Loss: 0.6708, Val Accuracy: 83.57%

Learning Rate: 0.00100

New best model saved with Validation Accuracy: 83.57%!

Epoch 2/2

```
Training: 100%|██████████| 53/53 [00:29<00:00, 1.81it/s, Loss=0.4471, Accuracy=90.65%]  
Validating: 100%|██████████| 14/14 [00:16<00:00, 1.19s/it]  
Train Loss: 0.4471, Train Accuracy: 90.65%  
Val Loss: 0.1523, Val Accuracy: 96.43%  
Learning Rate: 0.000100
```

New best model saved with Validation Accuracy: 96.43%!

Training completed in 96s
Best Validation Accuracy: 96.43%

Test training completed with validation acc: 96.43%